

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular & Supplementary Winter Examination-2023**

**Course: B. Tech.**

**Branch : AIDS**

**Semester : III**

**Subject Code & Name: BTAIES304 & Computer Architecture & Operating Systems**

**Max Marks: 60**

**Date: 09/01/2024**

**Duration: 3 Hr.**

***Instructions to the Students:***

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

|                                                                           | (Level/CO) | Marks     |
|---------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                               |            | <b>12</b> |
| A) Explain about address sequencing in control memory with neat diagrams? | L1/CO1     | 6         |
| B) What are the basic operations performed by the processor?              | L1/CO1     | 6         |
| C) Write briefly about computer fundamental system?                       | L1/CO1     | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                |            | <b>12</b> |
| A) Explain Contiguous Allocation.                                         | L2/CO2     | 6         |
| B) Describe Implementation of Page Table.                                 | L2/CO2     | 6         |
| C) Explain Second Chance Page Replacement Algorithm.                      | L2/CO2     | 6         |
| <b>Q. 3 Solve Any Two of the following.</b>                               |            | <b>12</b> |
| A) Explain role of interrupts in process state transitions.               | L3/CO3     | 6         |
| B) Describe USB with it's Protocol.                                       | L3/CO3     | 6         |
| C) Explain Parallel Processors.                                           | L3/CO3     | 6         |
| <b>Q.4 Solve Any Two of the following.</b>                                |            | <b>12</b> |
| A) Explain operating system structure.                                    | L4/CO4     | 6         |
| B) Describe Micro Kernel System Structure with it's modules.              | L4/CO4     | 6         |
| C) Compare between Process and Thread.                                    | L4/CO4     | 6         |
| <b>Q. 5 Solve Any Two of the following.</b>                               |            | <b>12</b> |
| A) Describe Dining-Philosophers Problem.                                  | L5/CO5     | 6         |
| B) Explain Monitor Implementations using Semaphores.                      | L5/CO5     | 6         |
| C) Write a note on Banker's Algorithm.                                    | L5/CO5     | 6         |

**\*\*\* End \*\*\***